

RAJ JAISWAL

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PROFESSIONAL SUMMARY

Aspiring Data Scientist and Machine Learning Engineer pursuing a Bachelor of Technology in Computer Science Engineering at Jaypee University. Proficient in Python, C++, SQL, and Machine Learning with hands-on experience developing end-to-end AI and machine learning applications.

EDUCATION

Jaypee University

B.Tech in Computer Science Engineering Expected 2027

TECHNICAL SKILLS

Languages: C, C++, SQL, Python

Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, FastAPI

Tools: Git, GitHub, Google Colab

PROJECTS

YouTube Video Chat Assistant

Code: [GitHub Repository](#)

Python, FastAPI, LangChain, FAISS, Google Gemini, Chrome Extension API

- Developed a Chrome Extension-based Retrieval-Augmented Generation (RAG) assistant that enables users to ask natural language questions about YouTube videos by grounding responses in video transcripts using LangChain, FAISS, and Google Gemini.
- Engineered an end-to-end semantic retrieval pipeline that extracts transcripts, performs chunking, generates embeddings, and retrieves relevant context from a FAISS vector store to improve answer relevance and reduce LLM hallucinations.
- Optimized the RAG backend by introducing transcript caching and chat history management, reducing repeated processing for previously queried videos and enabling context-aware follow-up questions.

Telecom Customer Churn Prediction Platform

Code: [GitHub Repository](#)

Python, LightGBM, Scikit-learn, SHAP, Streamlit, Pandas, NumPy, Google Gemini API

- Developed an end-to-end telecom customer churn prediction platform using LightGBM and Scikit-learn, achieving a 98% F1-score to identify at-risk customers through an interactive Streamlit application.
- Integrated SHAP Explainability with Google Gemini to transform feature-level model explanations into stakeholder-friendly business insights and actionable customer retention recommendations.
- Built a Customer Strategy Simulator that enables businesses to compare baseline and simulated churn predictions, supporting data-driven evaluation of customer retention strategies before deployment.

MovieFlix: AI-Powered Movie Recommender

Code: [GitHub Repository](#)

Python, FastAPI, Scikit-learn, Joblib, HTML, CSS, JavaScript

- Developed a full-stack movie recommendation system using FastAPI and Content-Based Filtering, delivering personalized recommendations from the TMDB 5000 dataset through a responsive web interface.
- Engineered an NLP recommendation pipeline by combining movie metadata (overview, genres, keywords, cast, and crew) with TF-IDF vectorization and Cosine Similarity to generate relevant movie recommendations and support fuzzy title search.
- Built and optimized REST APIs for movie search, recommendations, trending movies, and detailed metadata, while using Joblib for efficient model loading to enable low-latency real-time inference.

CERTIFICATIONS

- AI Foundations** – OpenAI Academy
- Agents and Workflows** – OpenAI Academy

ACHIEVEMENTS

- HackRide 2024** – Secured 25th Rank among 150+ participating teams.